

Cohomological invariants and depth

SGA 2, Exposé III

English draft

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About this draft

This is an unofficial English translation of Exposé III, “Cohomological invariants and depth”, from SGA 2. The source volume is A. Grothendieck (notes by a group of auditors), with an exposé by M. Raynaud, *Cohomologie locale des faisceaux cohérents et théorèmes de Lefschetz locaux et globaux* (SGA 2), Séminaire de géométrie algébrique du Bois Marie, 1962. It follows the slightly corrected SMF recomposition, arXiv:math/0511279.

This draft contains the assigned body of the source, including proofs, original footnotes, and editor notes (N.D.E.). Statement numbering and mathematical notation follow the source. References to material outside this draft retain their original numbers. Apparent mathematical misprints in the source have been retained; they are recorded in the accompanying README, separately from the translated text. Scholarly proofreading remains outstanding.

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EXPOSÉ III

Cohomological invariants and depth

1. Recollections

We shall state some definitions and results which the reader will find for example in Chapter I of the course given by J.-P. Serre at the Collège de France in 1957–58.¹

DEFINITION 1.1. Let A be a ring, (commutative with unit element as in all that follows) and let M be an A -module (unitary as in all that follows), one calls:

- annihilator of M and one denotes by $\text{Ann } M$ the set of $a \in A$ such that for every $m \in M$ one has $am = 0$.
- support of M and one denotes by $\text{Supp } M$ the set of prime ideals $\mathfrak{p}$ of A such that the localized module $M_{\mathfrak{p}}$ is nonzero.
- “assassin of M ” or “set of prime ideals associated with M ” and one denotes by $\text{Ass } M$ the set of prime ideals $\mathfrak{p}$ of A such that there exists a nonzero element of M whose annihilator is $\mathfrak{p}$.

If $\mathfrak{a}$ is an ideal of A , we shall denote by $\mathfrak{r}(\mathfrak{a})$ the radical of $\mathfrak{a}$ in A , i.e. the set of elements of A a power of which lies in $\mathfrak{a}$.

The following results are valid if one assumes that A is noetherian and M of finite type.

PROPOSITION 1.1. (i) $\text{Ass } M$ is a finite set.

(ii) For an element of A to annihilate a nonzero element of M , it is necessary and sufficient that it belong to one of the ideals associated with M .

(iii) The radical of the annihilator of M , $\mathfrak{r}(\text{Ann } M)$, is the intersection of the associated ideals of M which are minimal (for the inclusion relation in $\text{Ass } M$).

PROPOSITION 1.2. Let $\mathfrak{p}$ be a prime ideal of A ; the following assertions are equivalent:

- (i) $\mathfrak{p} \in \text{Supp } M$.
- (ii) There exists $\mathfrak{q} \in \text{Ass } M$ such that $\mathfrak{q} \subset \mathfrak{p}$.
- (iii) $\mathfrak{p} \supset \text{Ann } M$.
- (iii bis) $\mathfrak{p} \supset \mathfrak{r}(\text{Ann } M)$.

PROPOSITION 1.3. Let N be an A -module of finite type; one has the formula:

$$\text{Ass Hom}_A(N, M) = \text{Supp } N \cap \text{Ass } M.$$

2. Depth

Throughout this section, A denotes a commutative ring, I an ideal of A , M and N two A -modules. One will denote by X the prime spectrum of A (one will not use its structure sheaf in this section) and by Y the variety of I , $Y = \text{Supp}(A/I) = \{\mathfrak{p} \in X, \mathfrak{p} \supset I\}$.

LEMMA 2.1. Suppose that A is noetherian and that the modules M and N are of finite type. Suppose moreover that $\text{Supp } N = Y$. Then the following assertions are equivalent:

- (i) $\text{Hom}_A(N, M) = 0$.

¹**N.D.E.** the re-edition of Serre’s text (Serre J.-P., *Algèbre locale. Multiplicités*, Cours au Collège de France, 1957–1958, written by Pierre Gabriel, second edition, Lect. Notes in Math., vol. 11, Springer-Verlag, 1965) no longer contains the proofs of these statements. One may refer to (Bourbaki N., *Algèbre commutative*, Masson), as Serre himself suggests.

- (ii) $\text{Supp } N \cap \text{Ass } M = \emptyset$.
- (iii) *The ideal I is not a zero-divisor in M , which means that for every $m \in M$, $Im = 0$ implies $m = 0$.*
- (iv) *There exists in I an M -regular element. (An element a of A is said to be M -regular if the homothety of ratio a in M is injective.)*
- (v) *For every $\mathfrak{p} \in Y$, the maximal ideal $\mathfrak{p}A_{\mathfrak{p}}$ of the local ring $A_{\mathfrak{p}}$ is not associated with $M_{\mathfrak{p}}$. In a formula: $\mathfrak{p}A_{\mathfrak{p}} \notin \text{Ass } M_{\mathfrak{p}}$.*

PROOF. (i) $\Leftrightarrow$ (ii) since $\text{Ass Hom}_A(N, M) = \emptyset$ is equivalent to (ii) by proposition 1.3 and to (i) by an easy consequence of proposition 1.2.

(iii) $\Rightarrow$ (ii) by contradiction: “there exists $\mathfrak{p} \in \text{Supp } N \cap \text{Ass } M$ ” implies that $\mathfrak{p} \supset I$ and that there exists $m \in M$ whose annihilator is $\mathfrak{p}$, hence $Im \subset \mathfrak{p}m = 0$, which contradicts (iii).

(iv) $\Rightarrow$ (iii) trivially.

(ii) $\Leftrightarrow$ (iv) since $\text{Supp } N = Y$, hence (ii) means that I is not contained in any ideal associated with M , or equivalently (since the ideals associated with M are prime and finite in number), that I is not contained in the union of the ideals associated with M . Now, by proposition 1.1 (ii), this set is the set of elements of A which are not M -regular.

(i) $\Rightarrow$ (v); indeed, if $\text{Hom}_A(N, M) = 0$ and if $\mathfrak{p} \in Y$, one deduces, by virtue of the formula

$$(\text{Hom}_A(N, M))_{\mathfrak{p}} = \text{Hom}_{A_{\mathfrak{p}}}(N_{\mathfrak{p}}, M_{\mathfrak{p}}),$$

that $\text{Hom}_{A_{\mathfrak{p}}}(N_{\mathfrak{p}}, M_{\mathfrak{p}}) = 0$, hence, thanks to proposition 1.3,

$$\text{Supp } N_{\mathfrak{p}} \cap \text{Ass } M_{\mathfrak{p}} = \emptyset,$$

now $\mathfrak{p}A_{\mathfrak{p}} \in \text{Supp } N_{\mathfrak{p}}$, hence $\mathfrak{p}A_{\mathfrak{p}} \notin \text{Ass } M_{\mathfrak{p}}$.

(v) $\Rightarrow$ (i); indeed, if $\mathfrak{p} \in \text{Ass } M$, there exists $m \in M$ whose annihilator is $\mathfrak{p}$, hence the canonical image of m in $M_{\mathfrak{p}}$ is nonzero, hence its annihilator is an ideal which contains $\mathfrak{p}$, hence $\mathfrak{p}A_{\mathfrak{p}}$, hence is equal to it. The ideal $\mathfrak{p}A_{\mathfrak{p}}$ is therefore associated with $M_{\mathfrak{p}}$, hence $\mathfrak{p} \notin Y$ by (v), whence (i). $\square$

We shall work with these conditions, replacing the functor Hom by its derived functors.

THEOREM 2.2. *Let A be a commutative ring, I an ideal of A , M an A -module. Let n be an integer.*

- a) *If there exists a sequence $f_1, \dots, f_{n+1}$ of elements of I which forms an M -regular sequence (i.e. if f_1 is M -regular and if f_{i+1} is regular in $M/(f_1, \dots, f_i)M$ for $i \leq n$), then for every A -module N annihilated by a power of I , one has:*

$$\text{Ext}_A^i(N, M) = 0 \text{ for } i \leq n.$$

- b) *If moreover A is noetherian, if M is of finite type, and if there exists an A -module N of finite type such that $\text{Supp } N = V(I)$ and such that $\text{Ext}_A^i(N, M) = 0$ for $i \leq n$, then there exists a sequence $f_1, \dots, f_{n+1}$ of elements of I which is M -regular.*

Let us first prove a), by induction. If $n < 0$ the statement is empty.

If $n \geq 0$, suppose that a) has been proved for $n' < n$; by hypothesis there exists $f_1 \in I$ which is M -regular. Let us denote by f_1^i multiplication by f_1 in $\text{Ext}_A^i(N, M)$ and by f_1^M multiplication by f_1 in M . The sequence

$$(1) \quad 0 \longrightarrow M \xrightarrow{f_1^M} M \longrightarrow M/f_1M \longrightarrow 0$$

is exact, hence so is the sequence:

$$\text{Ext}_A^{i-1}(N, M/f_1M) \xrightarrow{\delta} \text{Ext}_A^i(N, M) \xrightarrow{f_1^i} \text{Ext}_A^i(N, M).$$

By hypothesis $I^n N = 0$, hence f_1^0 and nilpotent; Ext_A^i is a universal functor, and the same is true of f_1^i for every i . Moreover, there exists a regular sequence in M/f_1M which has n elements, hence, by the induction hypothesis,

$$\text{Ext}_A^{i-1}(N, M) = 0 \text{ if } i \leq n - 1.$$

One deduces that if $i \leq n$, f_1^i is both nilpotent and injective, hence $\text{Ext}_A^i(N, M) = 0$.

Let us prove b), likewise by induction. If $n < 0$, the statement is empty.

If $n = 0$, b) follows from the assertion (i) $\Rightarrow$ (iv) of lemma 2.1.

If $n > 0$, by b) for $n = 0$, there exists an element $f_1 \in I$ which is M -regular; from the exact sequence (1), one deduces the exact sequence:

$$(2) \quad \text{Ext}_A^{i-1}(N, M) \longrightarrow \text{Ext}_A^{i-1}(N, M/f_1M) \longrightarrow \text{Ext}_A^i(N, M).$$

One concludes that the hypotheses of b) are satisfied for the module M/f_1M and for the integer $n - 1$. By the induction hypothesis, there exists a sequence of n elements of I which is regular for M/f_1M , which implies that there exists a sequence of $n + 1$ elements of I , beginning with f_1 , and which is M -regular.

This theorem invites us to generalize in the following way the classical definition of the depth of a module of finite type over a noetherian ring:

DEFINITION 2.3. Let A be a commutative ring with unit element, let M be an A -module, let I be an ideal of A . One calls I -depth of M , and one denotes by $\text{prof}_I M$, the supremum in $\mathbf{N} \cup \{+\infty\}$ of the set of natural numbers n which are such that for every A -module of finite type N annihilated by a power of I , one has

$$\text{Ext}_A^i(N, M) = 0 \text{ for every } i < n.$$

One deduces from the preceding theorem that if n is the supremum of the lengths of the M -regular sequences of elements of I , one has $n \leq \text{prof}_I M$.

More precisely:

PROPOSITION 2.4. Let A be a commutative ring, I an ideal of A and let M be an A -module, and finally let $n \in \mathbf{N}$. Consider the assertions:

- (1) $n \leq \text{prof}_I M$.
- (2) For every A -module of finite type N which is annihilated by a power of I , one has:

$$\text{Ext}_A^i(N, M) = 0 \text{ for } i < n.$$

- (3) There exists an A -module of finite type N such that $\text{Supp } N = V(I)$ and such that $\text{Ext}_A^i(N, M) = 0$ if $i < n$.
- (4) There exists an M -regular sequence of length n formed of elements of I .

One has the following logical implications:

$$(1) \iff (2) \iff (4) \\ \Downarrow \\ (3)$$

Moreover if A is noetherian and M of finite type, these conditions are equivalent.

PROOF. (1) $\Leftrightarrow$ (2) by definition and (2) $\Rightarrow$ (3) upon taking $N = A/I$; Moreover (4) $\Rightarrow$ (2) by theorem 2.2 a). Finally, if A is noetherian and M of finite type, (3) $\Rightarrow$ (4) by theorem 2.2 b). $\square$

We assume A noetherian and M of finite type until the end of this section.

COROLLARY 2.5. Let $f \in I$ be an M -regular element; one has:

$$\text{prof}_I M = \text{prof}_I(M/fM) + 1.$$

Indeed, if $n \leq \text{prof}_I(M/fM)$, there exists a sequence of elements of I , $f_1, \dots, f_n$, which is (M/fM) -regular; hence the sequence $f, f_1, \dots, f_n$ is M -regular, hence $n + 1 \leq \text{prof}_I M$, hence $\text{prof}_I M \geq \text{prof}_I(M/fM) + 1$. On the other hand, by the exact sequence (2), if $i \leq \text{prof}_I M$, one has $\text{Ext}_A^{i-1}(N, M/fM) = 0$, hence $\text{prof}_I M - 1 \leq \text{prof}_I(M/fM)$.

COROLLARY 2.6. Every finite M -regular sequence, formed of elements of I , can be extended to a maximal M -regular sequence, whose length is necessarily equal to the I -depth of M .

REMARK 2.7. One can scarcely restrain oneself from saying that an A -module is all the more beautiful the greater its depth. A module whose support does not meet $V(I)$ is among the most beautiful; indeed, one can prove that for $\text{prof}_I M$ to be finite, it is necessary and sufficient that $\text{Supp } M \cap V(I) \neq \emptyset$.

REMARK 2.8. If A is a semi-local ring, let $\mathfrak{r}(A)$ be its radical and $k = A/\mathfrak{r}(A)$ its residue ring. The interesting notion of depth is obtained by taking for I the radical of A . We shall therefore agree to denote simply by $\text{prof } M$ the $\mathfrak{r}(A)$ -depth of an A -module M . In this case one recovers the notion of “homological codimension”, (cf. Serre, *op. cit.* note (1), page 1), which was denoted $\text{codh}_A M$, and which is defined as the infimum of the integers i such that $\text{Ext}_A^i(k, M) \neq 0$; indeed $\text{Supp } k = V(\mathfrak{r}(A))$.

PROPOSITION 2.9. *If A is noetherian and M of finite type, one has:*

$$\text{prof}_I M = \inf_{\mathfrak{p} \in V(I)} \text{prof } M_{\mathfrak{p}}.$$

COROLLARY 2.10. *If A is a noetherian semi-local ring, and if M is an A -module of finite type, one has:*

$$\text{prof } M = \inf_{\mathfrak{m}} \text{prof } M_{\mathfrak{m}},$$

where $\mathfrak{m}$ runs through the set of maximal ideals of A .

The corollary follows immediately from the proposition 2.9; indeed the prime ideals which contain the radical are the maximal ideals.

On the other hand, let $f \in I$; if f is M -regular, if $\mathfrak{p} \in X$ and if $\mathfrak{p} \supset I$, the image g of f in $A_{\mathfrak{p}}$ belongs to $\mathfrak{p}A_{\mathfrak{p}}$, the maximal ideal of $A_{\mathfrak{p}}$; moreover g is $M_{\mathfrak{p}}$ -regular, as follows from the exact sequence

$$(3) \quad 0 \longrightarrow M_{\mathfrak{p}} \xrightarrow{g'} M_{\mathfrak{p}} \longrightarrow (M/fM)_{\mathfrak{p}} \longrightarrow 0,$$

where g' denotes the homothety of ratio g in $M_{\mathfrak{p}}$. This exact sequence also gives that $(M/fM)_{\mathfrak{p}}$ is isomorphic to $M_{\mathfrak{p}}/gM_{\mathfrak{p}}$; applying corollary 2.5 to M and to $M_{\mathfrak{p}}$, one deduces, by induction, that $\text{prof}_I M \leq \nu(M)$, where one has set for every M :

$$\nu(M) = \inf_{\mathfrak{p} \in V(I)} \text{prof } M_{\mathfrak{p}}.$$

More precisely, still by induction, one knows, if f is M -regular, that $\nu(M) = \nu(M/fM) + 1$; it remains therefore to prove that if $\nu(M) \geq 1$, there exists an M -regular element in I . Now applying lemma 2.1 to $M_{\mathfrak{p}}$, $A_{\mathfrak{p}}$ and $\mathfrak{p}A_{\mathfrak{p}}$ for every $\mathfrak{p} \in V(I)$, one sees that $\mathfrak{p}A_{\mathfrak{p}} \notin \text{Ass } M_{\mathfrak{p}}$, hence, applying lemma 2.1 to A , M and I , one has the conclusion.

PROPOSITION 2.11. *Let $u: A \rightarrow B$ be a homomorphism of noetherian rings. Let I be an ideal of A , M an A -module of finite type. Set $I_B = I \otimes_A B$ and $M_B = M \otimes_A B$. If B is A -flat, one has:*

$$\text{prof}_{I_B} M_B \geq \text{prof}_I M;$$

moreover if B is faithfully flat over A , one has equality.

Indeed, let $N = A/I$; by flatness one has: $N \otimes_A B = B/I_B$; set $N_B = N \otimes_A B$. Still by flatness and noetherian hypotheses, one has:

$$\text{Ext}_B^i(N_B, M_B) = \text{Ext}_A^i(N, M) \otimes_A B,$$

hence $\text{Ext}_A^i(N, M) = 0$ implies $\text{Ext}_B^i(N_B, M_B) = 0$, and the converse is true if B is faithfully flat over A .

3. Depth and topological properties

LEMMA 3.1. *Let X be a topological space, Y a closed subspace, let F be a sheaf of abelian groups on X . Put $U = X - Y$. If n is an integer, the following conditions are equivalent:*

- (i) $\underline{H}_Y^i(X, F) = 0$ if $i < n$.
- (ii) *For every open set V of X , the group homomorphism*

$$H^i(V, F) \longrightarrow H^i(V \cap U, F)$$

is bijective if $i < n - 1$ and injective if $i = n - 1$.

- (iii) *For every open set V of X ,*

$$H_{Y \cap V}^i(V, F|_V) = 0 \text{ if } i < n.$$

PROOF. (ii) $\Leftrightarrow$ (iii); indeed, let V be an open set of X , put $X' = V$, $Y' = Y \cap V$, $F' = F|_V$, $U' = X' - Y'$; Y' is closed in X' , one therefore has an exact sequence:

$$H_{Y'}^i(X', F') \longrightarrow H^i(X', F') \xrightarrow{\rho_i} H^i(U', F') \longrightarrow H_{Y'}^{i+1}(X', F').$$

If the extreme terms are zero, the homomorphism ρ_i is bijective, and if the left-hand term is zero, ρ_i is injective. Hence (iii) $\Rightarrow$ (ii). Conversely, if $i < n$, $H_{Y'}^i(X', F')$ is zero because ρ_i is injective and ρ_{i-1} surjective.

- (i) $\Rightarrow$ (iii); indeed the spectral sequence “of passage from local to global” gives:

$$H^p(X, \underline{H}_Y^q(X, F)) \Longrightarrow H_Y^{p+q}(X, F).$$

Now, by hypothesis $\underline{H}_Y^q(X, F) = 0$ if $q < n$, hence $H_Y^{p+q}(X, F) = 0$ if $p + q < n$.

- (iii) $\Rightarrow$ (i); indeed (iii) expresses that the presheaf

$$V \longmapsto H_{Y \cap V}^i(V, F|_V)$$

is zero, hence also the associated sheaf which is $\underline{H}_Y^i(X, F)$, because Y is closed. $\square$

REMARK 3.2. The equivalence of (i) is (ii) has been proved in proposition ?? ??. As one remarked then, if $n \geq 2$, one may omit the condition that ρ_{n-1} be injective.²

PROPOSITION 3.3. *Let X be a locally noetherian prescheme, Y a closed subprescheme of X , F a coherent $\mathcal{O}_X$ -module. The conditions of lemma 3.1 are equivalent to each of the following conditions:*

- (iv) *For every $x \in Y$, one has $\text{prof } F_x \geq n$;*
- (v) *For every coherent $\mathcal{O}_X$ -module G on X , with support contained in Y one has*

$$\underline{\text{Ext}}_{\mathcal{O}_X}^i(G, F) = 0 \text{ if } i < n;$$

- (vi) *There exists a coherent $\mathcal{O}_X$ -module G whose support is equal to Y and such that*

$$\underline{\text{Ext}}_{\mathcal{O}_X}^i(G, F) = 0 \text{ if } i < n.$$

If X is affine, one has done everything needed (cf. proposition 2.4) to prove the equivalence of the three conditions of proposition 3.3; now they are local, apart from the implication (v) $\Rightarrow$ (vi), but one may then take $G = \mathcal{O}_Y$ and invoke again proposition 2.4. It therefore suffices to prove (i) $\Rightarrow$ (vi) and (v) $\Rightarrow$ (i).

Let J be the ideal of Y , it is a coherent sheaf of ideals; let $\mathcal{O}_{Y_m} = \mathcal{O}_X/J^{m+1}$, it is a coherent $\mathcal{O}_X$ -module whose support is equal to Y , and one knows (theorem ?? ??b) that

$$\underline{H}_Y^i(X, F) = \varinjlim_m \underline{\text{Ext}}_{\mathcal{O}_X}^i(\mathcal{O}_{Y_m}, F),$$

hence (v) $\Rightarrow$ (i). Moreover, the transition morphisms are epimorphisms in the projective system of the $\mathcal{O}_{Y_m}$.

If the functor $\underline{\text{Ext}}^i$ is left exact in its first argument, at least when the latter is in the category of coherent $\mathcal{O}_X$ -modules with support contained in Y , the transition morphisms of

²N.D.E. the original edition reproduced a proof, not entirely correct.

the inductive system obtained by applying $\underline{\text{Ext}}^i$ to the $\mathcal{O}_{Y_m}$ will be injective, now (i) implies that the limit is zero, hence (i) will imply that the modules $\underline{\text{Ext}}^i_{\mathcal{O}_X}(\mathcal{O}_{Y_m}, F)$ are zero for every m . Let us argue by induction. The statement is trivial for $n < 0$. Suppose that (i) $\Rightarrow$ (vi) for $n < q$, then (i) $\Rightarrow$ (v), hence, by the exact sequence of the $\underline{\text{Ext}}$, $\underline{\text{Ext}}^q$ is left exact in its first argument, hence the modules $\underline{\text{Ext}}^q_{\mathcal{O}_X}(\mathcal{O}_{Y_m}, F)$ are zero for every m . Hence (i) $\Rightarrow$ (vi) for $n \leq q$. $\square$

EXAMPLE 3.4. Let A be a noetherian local ring, $\mathfrak{m}$ its maximal ideal, M an A -module of finite type, and finally let n be an integer. Put $X = \text{Spec}(A)$, $Y = \{\mathfrak{m}\}$, $U = X - Y$. Let F be the sheaf associated to M . The following conditions are equivalent:

- 1) $\text{prof } M \geq n$;
- 2) the natural homomorphism

$$H^i(X, F) \longrightarrow H^i(U, F)$$

is injective if $i = n - 1$, bijective if $i < n - 1$;

- 3) $\text{Ext}_A^i(k, M) = 0$ if $i < n$, where $k = A/\mathfrak{m}$;
- 4) $H_Y^i(X, F) = 0$ if $i < n$.

Taking remark 3.2 into account, one obtains:

COROLLARY 3.5. *Let X be a locally noetherian prescheme, Y a closed subprescheme of X , F a coherent $\mathcal{O}_X$ -module; the following conditions are equivalent:*

- 1) *for every $x \in Y$, $\text{prof } F_x \geq 2$;*
- 2) *for every open set V of X , the natural homomorphism*

$$H^0(V, F) \longrightarrow H^0(V \cap (X - Y), F)$$

is bijective.

THEOREM 3.6 (Hartshorne). *Let X be a locally noetherian prescheme, Y a closed subprescheme of X . Suppose that, for every $x \in Y$, $\text{prof } \mathcal{O}_{X,x} \geq 2$; then the natural map*

$$\pi_0(X) \longrightarrow \pi_0(X - Y)$$

is bijective.

PROOF. Since X is locally noetherian, X is locally connected; it therefore suffices to prove that for X to be connected it is necessary and sufficient that $X - Y$ be so. Now, for a locally ringed space $(X, \mathcal{O}_X)$ to be connected, it is necessary and sufficient that $H^0(X, \mathcal{O}_X)$ not be a direct product of two non-zero rings. But the hypothesis implies, by corollary 3.5 applied to $F = \mathcal{O}_X$, that the homomorphism

$$H^0(X, \mathcal{O}_X) \longrightarrow H^0(X - Y, \mathcal{O}_X)$$

is an isomorphism, whence the conclusion. $\square$

COROLLARY 3.7. *Let X be a locally noetherian prescheme. Let d be an integer such that $\dim \mathcal{O}_{X,x} \geq d$ implies $\text{prof } \mathcal{O}_{X,x} \geq 2$. Then, if X is connected, X is connected in codimension $d - 1$, i.e. if X' and X'' are two irreducible components of X , there exists a sequence of irreducible components of X :*

$$X' = X_0, X_1, \dots, X_n = X''$$

such that for every i , $0 \leq i < n$, the codimension of $X_i \cap X_{i+1}$ in X is less than or equal to $d - 1$.

Let us first remark that if X is Cohen–Macaulay, $d = 2$ will enjoy the property evoked above. In this connection, let us recall that one defines the codimension of Y in X as the infimum of the dimensions of the local rings in X of the points of Y .

PROOF. Let $\mathcal{F}$ be the set of closed subsets of X whose codimension in X is greater than or equal to d . One will note that $\mathcal{F}$ is an antifilter of closed subsets of X . Moreover, for a closed subset Y of X to be an element of $\mathcal{F}$, it is necessary and sufficient that, for every $y \in Y$ there exist an open neighbourhood V of X such that $Y \cap V$ be of codimension $\geq d$ in V . Finally, if X is connected and if $Y \in \mathcal{F}$, $X - Y$ is connected by Hartshorne's theorem. The corollary therefore follows from the following lemma, which is of a purely topological nature.

LEMMA 3.8. *Let X be a connected and locally noetherian topological space, and let $\mathcal{F}$ be an antifilter of closed subsets of X . One supposes that every closed subset $Y \subset X$ which belongs locally to $\mathcal{F}$, (i.e. for every point $x \in X$ there exists an open neighbourhood V of x in X and a $Y' \in \mathcal{F}$ such that $V \cap Y = V \cap Y'$), belongs to $\mathcal{F}$. The following conditions are equivalent:*

- (i) *for every $Y \in \mathcal{F}$, $X - Y$ is connected;*
- (ii) *if X' and X'' are two distinct irreducible components of X , there exists a sequence of irreducible components of X , $X_0, X_1, \dots, X_n$, such that $X' = X_0$, $X'' = X_n$ and, for every i , $1 \leq i < n$, $X_i \cap X_{i+1} \notin \mathcal{F}$.*

(ii) $\Rightarrow$ (i). Let $Y \in \mathcal{F}$, one must prove that the open set $U = X - Y$ is connected. Now, if U' and U'' are two irreducible components of U , there exist two irreducible components X' and X'' of X such that $X'' \cap U = U''$ and $X' \cap U = U'$; let $X_0, \dots, X_n$ be a sequence of irreducible components of X possessing the property evoked above; if one puts $U_i = X_i \cap U$, $0 \leq i \leq n$, the U_i will be irreducible components of U , moreover $U_i \cap U_{i+1}$ is non-empty if $0 \leq i < n$, for otherwise, $X_i \cap X_{i+1} \subset Y$ would be an element of $\mathcal{F}$, which is contrary to the choice of the sequence of the X_i . This implies that U is connected.

(i) $\Rightarrow$ (ii). Let $Y = \bigcup_{X', X''} X' \cap X''$ where one requires that X' and X'' be two *distinct* irreducible components of X such that $X' \cap X'' \in \mathcal{F}$. The family of the $X' \cap X''$ is locally finite because X is locally noetherian, moreover the $X' \cap X''$ are closed, hence Y is closed. Moreover, Y belongs locally to $\mathcal{F}$, hence $Y \in \mathcal{F}$. Hence $U = X - Y$ is connected. Let X' and X'' be two distinct irreducible components of X , let U' and U'' be their traces on U , which are non-empty by construction of Y . They are irreducible components of U , now U is connected, hence U being locally noetherian, there exists a sequence of irreducible components $U_0, \dots, U_n$ of U , such that $U_0 = U'$, $U_n = U''$ and $U_i \cap U_{i+1} \neq \emptyset$ and $U_i \neq U_{i+1}$, $0 \leq i < n$. Let $X_0, \dots, X_n$ be the sequence of irreducible components of X which is such that $X_i \cap U = U_i$; if $X_i \cap X_{i+1} \in \mathcal{F}$, by construction of $\mathcal{F}$, $U_i \cap U_{i+1} = \emptyset$ or $U_i = U_{i+1}$ which is not possible according to the choice of the U_i . $\square$

COROLLARY 3.9. *Let A be a noetherian local ring. One supposes that for every prime ideal $\mathfrak{p}$ of A , one has:*

$$(\dim A_{\mathfrak{p}} \geq 2) \implies (\text{prof } A_{\mathfrak{p}} \geq 2)$$

One supposes moreover that A satisfies the chain condition³. Then, for every $\mathfrak{p}$, minimal prime ideal of A , $\dim A/\mathfrak{p} = \dim A$, or again, all the irreducible components of $\text{Spec } A$ have the same dimension: that of A .

If X' and X'' are two irreducible components of X , one joins them by a chain having the properties enumerated in corollary 3.7; it then suffices to prove that two successive components have the same dimension, which follows from the second hypothesis.

EXAMPLE 3.10. Let X be the union of two complementary vector subspaces of respective dimensions 2 and 3 in a vector space of dimension 5; more precisely, let $X = \text{Spec } A$, with $A = B/\mathfrak{p} \cap \mathfrak{q}$, where $B = k[X_1, \dots, X_5]$, $\mathfrak{p}$ is the ideal generated by X_1, X_2, X_3 and $\mathfrak{q}$ the ideal generated by X_4 and X_5 ; X can be disconnected by the intersection point x of the two vector subspaces, hence the depth of $\mathcal{O}_{X,x}$ is equal to 1, for it cannot be ≥ 2 by virtue of

³Cf. EGA 0_{IV} 14.3.2

theorem 3.6. Another reason: the equidimensionality conclusion of the preceding corollary fails.

More generally, taking a union X of two vector subspaces of dimension $p, q \geq 2$ in a vector space of dimension $p+q$, for no embedding of X in a regular scheme is X even set-theoretically a complete intersection at the origin, for (up to modifying it without changing the underlying topological space in a neighbourhood of the origin), X would be Cohen–Macaulay hence of depth ≥ 2 at the origin, which is not the case.

REMARK 3.11. Let X be a locally noetherian prescheme, Y a closed subprescheme of X , F an $\mathcal{O}_X$ -module. Depth is a purely topological notion which is expressed in terms of vanishing of the $\underline{H}_Y^i(X, F)$ for $i < n$. One also wishes to study these sheaves for a given i , or for $i > n$. One proves in this connection the following result:

LEMMA 3.12. *Let m be an integer, for $\underline{H}_Y^i(X, F) = 0$, $i > m$ for every coherent F , it is necessary and sufficient that this be true for $F = \mathcal{O}_X$.*

By inductive limit it is then true for every quasi-coherent sheaf. For example, if Y can be described locally by m equations, or, as one says, if Y is locally set-theoretically a complete intersection (which occurs for example if X and Y are *non-singular*) it follows from the computation of the $\underline{H}_Y^i(X, F)$ by the Koszul complex that these sheaves are zero for $i > m$. Besides, one has used this fact implicitly in example 3.10. This cohomological condition is however not sufficient, as the following example proves:

EXAMPLE 3.13. Let $X = \text{Spec}(A)$, where A is a noetherian local ring, normal of dimension 2. Let Y be a curve in X . One can prove that the complement of the curve is an affine open set hence⁴ $\underline{H}_Y^i(\mathcal{O}_X) \approx \underline{H}_{X-Y}^{i-1}(\mathcal{O}_X) = 0$ for $i > 1$ because $H^{i-1}(X - Y, \mathcal{O}_X) = 0$. However one can construct a curve which is not described by an equation.

We shall seek⁵ conditions for the $\underline{H}_Y^i(X, F)$ to be coherent for a given i , which is not the case in general, as is shown by obvious examples, for example $H_m^n(A)$ for A a noetherian local ring of dimension $n > 0$; when for example A is a discrete valuation ring with field of fractions K , one finds $H_m^1(A) \simeq K/A$, which is not a module of finite type over A . To enlighten the reader, let us say that the problem posed is equivalent to the following: let $f: U \rightarrow X$ be an open immersion, let G be a coherent sheaf on U , find criteria for the higher direct images $R^i f_* G$ to be coherent sheaves on X for a given i . These conditions are necessary for the use of formal geometry that we have seen in exposé ?? and the following ones.

⁴N.D.E. There was a misprint in the original edition.

⁵Cf. Exp ??.